

week 7 //

1. $\lim_{x \rightarrow 6} (x-2)^2$
2. $\lim_{x \rightarrow 7} (10-x)^4$
3. $\lim_{t \rightarrow 4} \sqrt{t+2}$
4. $\lim_{y \rightarrow 4} 3|y-1|$
5. $\lim_{t \rightarrow -2} \frac{t+2}{t^2-4}$
6. $\lim_{t \rightarrow 3} \frac{t^2-9}{t-3}$
7. $\lim_{x \rightarrow 4} \frac{\sqrt{x-3}-1}{x-4}$
8. $\lim_{x \rightarrow 0} \frac{\sqrt{4+x}-2}{x}$

$$\begin{aligned}
 1. &= x^2 - 4x + 4 \\
 &= 36 - 24 + 4 \\
 &= 16 \\
 2 &= (10-7)^4 \\
 &= 3^4 \\
 &= 81 \\
 3. &= \sqrt{4+2} \\
 &= \sqrt{6}
 \end{aligned}$$

$$\begin{aligned}
 4. &\begin{cases} 3(y-1), & x \geq 0 \\ -3(y-1), & x < 0 \end{cases} \\
 &3(y-1); = 3(3) \\
 &= 9 \\
 &-3(y-1); = -3(3) \\
 &= -9 \\
 5. &= \frac{t+2}{t^2-2^2} = \frac{\cancel{t+2}}{(\cancel{t+2})(t-2)} = \frac{1}{t-2} \\
 &= \frac{1}{-4}
 \end{aligned}$$

$$6. = \frac{t^2 - 3^2}{t - 3} = \frac{(\cancel{t-3})(t+3)}{\cancel{t-3}} = t+3 = 6$$

$$\begin{aligned}
 8. &\frac{\sqrt{4+x}-2}{x} \times \frac{\sqrt{4+x}+2}{\sqrt{4+x}+2} \\
 &= \frac{(4+x) + 2\sqrt{4+x} - 2\sqrt{4+x} - 4}{x(\sqrt{4+x}+2)} \\
 &= \frac{\cancel{4} + x - \cancel{4}}{\cancel{x}(\sqrt{4+x}) + 2} = \frac{1}{\sqrt{4+x}+2} = \frac{1}{\sqrt{4+2}} \\
 &= \frac{1}{4}
 \end{aligned}$$

$$\begin{aligned}
 7. &\frac{\sqrt{x-3}-1}{x-4} \times \frac{\sqrt{x-3}+1}{\sqrt{x-3}+1} \\
 &= \frac{(x-3) + \sqrt{x-3} - \sqrt{x-3} - 1}{(x-4)(\sqrt{x-3}+1)} \\
 &= \frac{\cancel{x-4}}{\cancel{(x-4)}(\sqrt{x-3}+1)} \\
 &= \frac{1}{\sqrt{x-3}+1} \\
 &= \frac{1}{\sqrt{4-3}+1} = \frac{1}{\sqrt{1}+1} \\
 &= \frac{1}{2}
 \end{aligned}$$

$$\begin{aligned}
 &\frac{x-3}{\sqrt{x}-\sqrt{3}} \times \frac{\sqrt{x}+\sqrt{3}}{\sqrt{x}+\sqrt{3}} \\
 &= \frac{(x-3)(\sqrt{x}+\sqrt{3})}{x + \cancel{\sqrt{x}\sqrt{3}} - \cancel{\sqrt{x}\sqrt{3}} - 3} \\
 &= \frac{(\cancel{x-3})(\sqrt{x}+\sqrt{3})}{(\cancel{x-3})}
 \end{aligned}$$

$$\begin{aligned}
 &\lim_{x \rightarrow 0} \frac{\sqrt{x+1}-1}{x} \\
 &= \frac{\sqrt{x+1}-1}{x} \times \frac{\sqrt{x+1}+1}{\sqrt{x+1}+1} \\
 &= \frac{(x+1) - \cancel{\sqrt{x+1}} + \cancel{\sqrt{x+1}} - 1}{x(\sqrt{x+1}+1)} \\
 &= \frac{x}{x(\sqrt{x+1}+1)} = \frac{1}{\sqrt{x+1}+1} = \frac{1}{\sqrt{1}+1} = \frac{1}{2}
 \end{aligned}$$

$$\lim_{x \rightarrow -3} \frac{x^2 + x - 6}{x + 3} = \frac{(\cancel{x+3})(x-2)}{\cancel{x+3}} = x-2$$

$$= -3-2 = -5$$

$$\lim_{x \rightarrow 1} \frac{x^3 - 1}{x - 1} = \frac{(\cancel{x-1})(x^2 + x + 1)}{\cancel{x-1}}$$

$$= 1 + 1 + 1 = 3$$

	30	45	60
sin	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$
cos	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$
tan	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$

$$a^3 - b^3 = (a - b)(a^2 + ab + b^2)$$

$$a^3 + b^3 = (a + b)(a^2 - ab + b^2)$$

$$1. g(x) = (x^3 + 7x)(x + 3)$$

$$2. f(x) = \frac{6x-5}{x^2+1}$$

$$3. y = \frac{\sin x}{x^4}$$

$$1. g'(x) = x^4 + 3x^3 + 7x^2 + 21x$$

$$= 4x^3 + 9x^2 + 14x + 21$$

week 2 //

$$2. f'(x) = \frac{(x^2+1)(6) - (6x-5)(2x)}{(x^2+1)^2}$$

$$= \frac{6x^2 + 6 - 12x^2 + 10x}{x^4 + 2x^2 + 1}$$

$$= \frac{-6x^2 + 10x + 6}{x^4 + 2x^2 + 1}$$

$$3. y' = \frac{\cos x (x^4) - \sin x (4x^3)}{x^8}$$

$$= \frac{x^3 [(\cos x)(x) - 4(\sin x)]}{x^8}$$

$$= \frac{(\cos x)(x) - 4\sin x}{x^5}$$

$$f'(x) = 3x^2 + 2$$

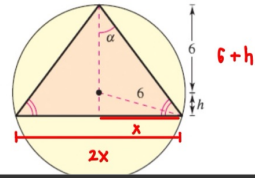
$$f'(x) = x^{\frac{1}{2}}$$

$$= \frac{1}{2} x^{\frac{1-2}{2}}$$

$$= \frac{1}{2} x^{-\frac{1}{2}}$$

week 4

จงคำนวณหาพื้นที่ของสามเหลี่ยมหน้าจั่วที่ใหญ่ที่สุด ที่สามารถบรรจุ
อยู่ในวงกลมรัศมี 6 หน่วย (ตามรูป) โดยในการคำนวณ ให้ใช้ฟังก์ชัน
ในการคำนวณพื้นที่ ที่มีตัวแปร h เป็นตัวแปรอิสระ



อยู่ใน $[1, 6]$

$$b = 2x \quad s = (6+h)$$

$$\begin{aligned} \text{area} &= \frac{1}{2} \times b \times s \\ &= \frac{1}{2} \times 2x \times (6+h) \end{aligned}$$

$$= x \times (6+h)$$

$$\text{หา } x, \quad 6^2 = x^2 + h^2$$

$$-x^2 = h^2 - 6^2$$

$$x^2 = 6^2 - h^2$$

$$x = \sqrt{36 - h^2}$$

$$\text{area} = \sqrt{36 - h^2} \times (6+h)$$

$$= \frac{(36 - h^2)^{\frac{1}{2}} - 2h}{\sqrt{36 - h^2}} (6+h)$$

$$= \sqrt{36 - h^2} + \frac{-16h - 2h^2}{\sqrt{36 - h^2}}$$

$$0 = \frac{36 - h^2 - 16h - 2h^2}{\sqrt{36 - h^2}}$$

$$0 = \frac{-3h^2 - 16h + 36}{\sqrt{36 - h^2}}$$